

Michael Capriotti

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Education

Northwestern University - Bachelor's degree in **Computer Science and Mathematics**, GPA: 3.92/4.0

Evanston, IL

- **Relevant Coursework:** Linear Algebra, Multivariable Calculus, Data Structures and Algorithms, Operating Systems, Design and Analysis of Algorithms, Discrete Mathematics, Human Computer Interaction

2024 — 2028

Relevant Work Experience

NASA Marshall Space Flight Center — Huntsville, AL

June — August 2026

Software Engineering Intern

- Developed a PostgreSQL database comprising ~ 40 tables for mass properties data, using a normalized schema that separates hierarchical vehicle structure from item data, allowing new revisions to efficiently reuse existing items while maintaining immutable history tracking; optimized query performance with targeted indexes and enforced data integrity through triggers and constraints.
- Built Python data-processing scripts to load 4M+ historical mass property values through a standardized file format that separates vendor-specific parsing from database logic, allowing new vendor formats to be integrated without modifying the database pipeline.
- Created a FastAPI web application for searching, hierarchy navigation, revision management, and import/export; validated database operations, file transformations, and data-processing workflows with 200+ unit and end-to-end tests, achieving $\sim 90\%$ coverage.

Northwestern Computer Science Department — Evanston, IL

January — June 2026

Undergraduate Teaching Assistant

- Supported a computer systems course with 150+ students through tutoring, exam proctoring, grading, and helping students with data representation, assembly, stack-based memory, virtual memory, caches, file systems, concurrency, and process scheduling.

Research Projects

Quantum Computing Research at Los Alamos National Laboratory — *Python, NumPy, SciPy, Matplotlib, Qiskit, CVXPY*

- Applied the Quantum Approximate Optimization Algorithm (QAOA) to solve combinatorial optimization problems (portfolio optimization, traveling salesman, and maximum independent set) by transforming QUBO problems into Max-Cut instances.
- Utilized semidefinite programming (SDP)-based “warm-start” strategies to leverage classical optimization for improved QAOA initialization and faster convergence. Submitted a revised [manuscript](#) [✉](#) to ACM Transactions on Quantum Computing.

Data Science at Northwestern Kellogg School of Management — *Python, Pytesseract, OCR, LLMs*

- Developed a scalable automated obituary classification pipeline to assess executive importance, transitioning from manual labeling (500 samples) to OCR extraction and LLM-based labeling. Presented work at [IMSAlloquium](#) [✉](#).

Software Projects

Stock Forecast — *Python, TensorFlow, Pandas, FastAPI, React, TypeScript, SQLite*

- Built a machine learning web application predicting next-day closing prices for 400+ companies using LSTM models trained on 5 years of historical market data, achieving an average mean absolute error (MAE) of 2.76%, with interactive visualizations.

Golf Swing Analyzer — *Python, Scikit-learn, Random Forest, MediaPipe, OpenCV, Flask*

- Engineered and deployed a full-stack web application for golf swing analysis, processing 100 labeled swings and extracting normalized 3D coordinates from 13 key body landmarks to classify videos as Pro or Amateur with a Random Forest model achieving $\sim 80\%$ accuracy; enabled video upload, trimming, and interactive pose annotations while minimizing backend memory usage.

Trading Simulation — *C++, STL*

- Created a single-ticker market simulator with a price-sorted order book and matching engine, implementing four trading strategies (Value Investor, Trend Follower, Stop Loss, Market Maker) and simulating execution and portfolio performance across 2,000 ticks.

Evolutionary Flappy Bird — *C#, Unity, Evolutionary Algorithm*

- Implemented a custom NEAT-inspired neural network in C# with elitism, mutation, and multi-cycle fitness evaluation, enabling progressive AI improvement across generations and automatic preservation of the best-performing models for replay.

Backend Developer — **Northwestern DISC** — *JavaScript, Node.js, PostgreSQL, AWS (Lambda, RDS), Firebase*

- Developed the database and backend APIs for a donation tracking platform used by a nonprofit, enabling secure role-based access, Excel data uploads, validation, filtering, and donation history queries for a frontend dashboard.

Technical Skills

Languages: Python, C/C++, JavaScript, TypeScript, C#, SQL, HTML/CSS.

Frameworks/Platforms: FastAPI, React, Node.js, Flask, AWS, Docker, GitHub Actions, Unity.

Libraries: NumPy, Pandas, SciPy, CVXPY, Scikit-learn, TensorFlow, Qiskit, Matplotlib, MediaPipe, Pytesseract

Additional Information

Programs: MIT Introduction to Technology, Engineering, and Science (MITES).

Activities: Develop & Innovate for Social Change (DISC), Consultants Advising Student Enterprises, Table Tennis Club.